

Anutej Sachin Kardele

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EDUCATION

Master of Science in Computer Science,
University at Buffalo, The State University of New York, Buffalo, NY
Bachelor of Technology in Computer Science,
Vellore Institute of Technology, India

Aug 2025 - Dec 2026
GPA: 3.75 / 4.0
Jul 2018 - May 2022
CGPA: 7.72 / 10.0

EXPERIENCE

Software Engineering Practicum, HeinOnline, Buffalo, NY

Jan 2026 – May 2026

- Engineered a Python-powered OCR and AI extraction pipeline leveraging Qwen language models, performing ablation studies to optimize multi-document processing; achieved a 90.58% extraction accuracy across 28,167 journal pages.
- Built services to cross-reference extracted PDF metadata against CrossRef APIs, mapping verified author records and publication metrics (ISSN, date, volume) into PostgreSQL databases to automate Flipper production ingestion.

Software Engineer, Simbott Technologies, Mumbai, India

Mar 2024 - Jul 2025

- Architected a scalable, component-based C# architecture for enterprise applications, using Git to reduce duplicated code by 80% and enable custom module implementation in under 5 minutes within secure environments.
- Co-engineered 3 custom peripherals in collaboration with the electronics team, authoring high-performance firmware to establish low-latency serial communication pipelines between edge-hardware and the central simulation engine.
- Engineered real-time telemetry processing pipelines for custom physical controllers, optimizing sensor data streams to decrease average system latency by 68% in air-gapped environments.

Software Developer, Tecknotrove System Pvt Ltd, Mumbai, India

Jun 2022 - Nov 2023

- Implemented core C# application logic for enterprise training software, integrating a MySQL relational database to securely capture, store, and query thousands of user evaluation data points in real-time.
- Formulated a spatial-coordinate algorithm using vector raycasting for real-time event tracking, reducing computational workload by 82% (CPU utilization improvement) per interaction; bypassed legacy physics engines to eliminate boundary clipping errors.
- Executed a full-scale core engine migration (Unity 5.6 to 2019) for a legacy enterprise project, resolving 54 critical package dependencies and establishing strict hardware-to-software interfacing protocols.

SKILLS

- Languages:** C#, Java, Python, Go (Golang), JavaScript, C, HTML/CSS
- Databases & Infrastructure:** PostgreSQL, Docker, MySQL, Linux
- Frameworks:** React, Spring Boot, Pandas, NumPy, PyTorch
- Tools:** Git, Ollama, Auth0, Arduino IDE, Figma

PROJECT EXPERIENCE

OpenStream Microblogging Platform | Java, Spring Boot, PostgreSQL, JUnit 5, Docker

Sep 2026 – Present

- Engineered a Spring Boot REST API with 10 endpoints in a controller-service-repository architecture, mapping 3 entities to PostgreSQL via Spring Data JPA and Hibernate with DTOs separating persistence from API contracts.
- Modeled a self-referential many-to-many follow graph as an explicit join entity, enforcing referential integrity with composite unique constraints and foreign keys; eliminated N+1 lazy-loading by rewriting chronological feed retrieval as a single JPQL query with JOIN FETCH.
- Validated business rules and error paths with 14 JUnit 5 and Mockito unit tests, mocking repositories via constructor injection to run in under one second without a database; containerized with a multi-stage Docker build.

Interactive VS Code Portfolio & AI Terminal | React.js, REST APIs, GenAI, VFS

Feb 2026

- Designed a React SPA simulating a VS Code environment, implementing state management and an in-memory Virtual File System (VFS) to support modular windows, active tabs, and runtime component rendering.
- Engineered a custom lexical parser using tree traversal to process UNIX-style commands (ls, cd), and integrated a generative AI agent via RESTful APIs for context-aware, natural language user queries on professional experience.
- Automated deployment pipelines using GitHub Actions to continuously deliver React build artifacts to GitHub Pages.

Industrial Welding Simulator | C#, Arduino, Real-Time Data Pipelines

Dec 2024 – Jun 2025

- Engineered a low-latency simulation engine to process real-time spatial hardware inputs with sub-20ms synchronization, accelerating student certification rates by 87%.
- Architected an automated data pipeline to ingest and analyze high-frequency telemetry across 14 parameters (including work angles and CTWD), generating dynamic performance reports that saved an estimated \$2,500 in physical consumables per training cohort.

HONORS & AWARDS

National Winner, Smart India Hackathon 2020 | VR Fire Safety Simulator

Dec 2020